Statistical Learning and Machine Learning

Exercises (Week 3)

Book Exercises: 2.1, 2.5, 2.34, 2.35, 2.6 (optional)

Additional Exercises:

1. Prove eq. (2.7) by minimizing the log-likelihood function given in eq. (2.6) in the textbook.
2. Prove eq. (2.27) and (2.28) in the textbook.

Reading Exercise: Go through the proofs in Section 2.3, 2.3.1 and 2.3.2. Understand the eqs (2.94) - (2.98)

Statistical Learning and Machine Learning

Exercise Session (Week 3)

TOPIC 1

You are given $N = 6$ data points, $\{x_i\}_{i=1}^{6}$, the values of which are shown in the Figure 1. The data points reside in the 1D space i.e., $x_i \in \mathbb{R}$. It is further known that these points are independent and identically distributed according to the Gaussian distribution. You are asked to estimate the parameters of that Gaussian distribution from the above data points by using the maximum likelihood (ML) approach and the Bayesian approach. Please answer the following questions.

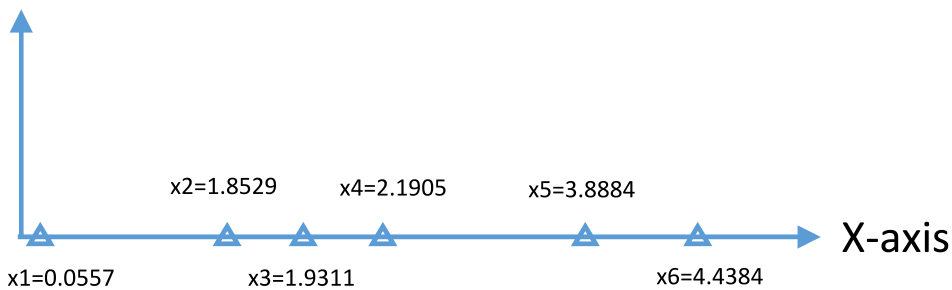

Fig. 1. Data points $\{x_i\}_{i=1}^{6}$ residing in the 1D space obtained from a Gaussian distribution with unknown mean μ and variance σ^2.

1) Estimate the mean of the Gaussian distribution, μ_{ML}, from the available data points by using the maximum likelihood (ML) method. (Score: 15%)
2) Estimate the variance of the Gaussian distribution. σ^2_{ML}, from the available data points by using the maximum likelihood (ML) method. (Score: 15%)
3) Comment on whether the above estimates of the mean μ_{ML} and the variance σ^2_{ML} are biased or unbiased? (Score: 10%)
4) If one or both the above estimates obtained in the previous step are biased, how would you make those unbiased? (Score: 10%)
5) Now, assume that the actual variance of the Gaussian distribution that led to the data points in Fig. 1 is known and is $\sigma^2 = 1$. Using that information and the same data points as in Fig. 1, estimate the mean of the Gaussian distribution μ_{Bayes} through the Bayesian approach by using the following conjugate form for the prior distribution of mean: $p(\mu) = \mathcal{N}(\mu|\mu_0 = 4, \sigma_0^2 = 3)$. (Score: 25%)
6) Given that the information about the prior density $p(\mu)$ is reasonable, which of the above density estimation approaches (i.e., ML or Bayesian) do you expect will perform well in the following settings: a) for smaller values of N (e.g., for $N = 6$ as in Fig. 1); b) for $N \to \infty$. (Score: 10%)
7) Briefly mention the reasons behind your answers in the previous question (6). (Score: 15%)